IGOR Final Report

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Executive Summary

This research initiative was launched to develop a mechanistic understanding of protein yield variations observed in Round 2 PURE system experiments. The IGOR process revealed that yield is not governed by optimizing individual components, but by complex, non-linear interactions within the core energy and catalysis subsystem. The process systematically evaluated several hypotheses, culminating in two highly-ranked, evolved theories. These top hypotheses propose that maximal yield occurs along a non-linear performance 'ridge' defined by either a dynamic 'kinetic flux balance' between ATP regeneration and consumption, or a 'hydrotropically-stabilized phase boundary' where high NTP concentrations prevent component precipitation. A multivariate experimental plan has been designed to test these theories by mapping this multi-dimensional interaction space.

Top-Ranked Hypotheses

1: Hypothesis on the Kinetic Flux Balance of Energy Metabolism and Magnesium Buffering

Hypothesis: Building upon the winning 'stoichiometric balance' concept, this hypothesis refines the mechanism from a static initial ratio to a dynamic, rate-based equilibrium. Maximal yield is achieved not by a specific initial concentration of Mg, CP, and ATP, but by maintaining a kinetic flux balance where the rate of Mg-dependent ATP regeneration from the CP/CK system precisely matches the rate of Mg-ATP consumption by transcription and translation. This balance acts as a homeostatic mechanism that buffers the concentration of free Mg^{2+}, the system's central regulated variable. The observed bimodal performance 'ridge' represents two distinct domains of this kinetic equilibrium (e.g., high-regeneration/high-consumption and medium-regeneration/medium-consumption). Points between these domains represent a kinetic mismatch, leading to failure modes: either rapid phosphate accumulation and Mg^{2+} precipitation when regeneration outpaces consumption, or energy depletion when consumption outpaces regeneration. **Final Rationale:** This hypothesis evolves the top-ranked 'Architect 1' by incorporating insights from 'Quant 1' and addressing the static nature of the original. The evolution strategy is HYPOTHESIS_COMBINATION. It strengthens the winning concept by shifting the frame from a static 'stoichiometric ratio' to a more mechanistically precise 'kinetic flux balance'. This provides a more powerful explanation for the observed experimental data, particularly the failure of conditions *between* the two high-yield optima noted in the Round 1 results, which are explained here as regions of kinetic mismatch. **Key Supporting Evidence:** - Debate Insight: The superiority of 'Architect 1' was its direct explanatory power for the bimodal Mg-CP performance ridge. - Pontes et al., 2015: High ATP chelates free Mg2+, creating a need for a buffered, not just high, concentration. - Kim & Swartz, 1999: Phosphate byproduct from the CP/CK system is inhibitory, primarily by sequestering Mg2+, highlighting the danger of an overactive regeneration flux. - Saks et al., 1975: Creatine Kinase activity is Mg2+-dependent, confirming that the regeneration rate itself is coupled to the availability of free Mg2+.

2: Hypothesis of Yield Optimization at a Hydrotropically-Stabilized Phase Boundary

Hypothesis: This hypothesis integrates the concepts of a multi-dimensional performance boundary (Quant 1) and the secondary functions of NTPs (Explorer 1). It posits that the PURE system operates as a concentrated, near-saturated solution where maximal yield occurs at a critical phase boundary, just before the precipitation of key components like magnesium phosphate. The non-linear performance 'ridge' observed in experiments is the mapping of this phase boundary. We hypothesize that NTPs (ATP and GTP) play a crucial dual role: 1) as energy currency and 2) as hydrotropes that prevent protein/salt aggregation, effectively expanding the soluble operating window. Therefore, optimization is a trade-off: high NTP concentrations are required to push the phase boundary and keep components soluble at the high concentrations needed for maximal reaction rates, but this same increase in NTP concentration exacerbates the chelation of free Mg2+. The optimal state is a co-limited one, where NTP hydrotropy provides just enough stabilization to operate at the very edge of precipitation, maximizing both component density and catalytic activity. **Final Rationale:** This hypothesis is an evolution via NOVELTY_INCREASE and HYPOTHESIS_COMBINATION, merging the strongest elements of all top hypotheses. It addresses the weakness of 'Explorer 1' (being disconnected from the primary data) by directly linking the novel hydrotropy concept to the empirically observed 'performance ridge'. It reframes the ridge not just as a stoichiometric balance but as a physical phase boundary, providing a new, testable, and powerful mechanistic explanation for the sharp performance cliffs seen in the data. **Key Supporting Evidence:** - Debate Insight: ‘Quant 1’ was praised for its concept of a ‘multi-dimensional boundary’ or ‘operational ridge’ that directly explained the Round 1 data. - Identified Literature Gap: ‘The dual role of ATP/GTP as both energy sources and hydrotropes… and how this… trades off against… Mg2+ chelation… is not well understood.’ - Zemella et al., 2015: Highlights the inhibitory effect of phosphate accumulation, a key driver of precipitation in the PURE system. - Analogy: This mirrors principles in materials science and chemistry where optimal processes often occur at phase transitions or boundaries, representing a state of maximum potential before systemic collapse.

Proposed Experimental Plan

Objective

To experimentally map the multi-dimensional interaction landscape of the PURE system’s core energy and catalysis components ([Magnesium acetate], [Creatine phosphate], [ATP], [GTP]) to validate the top-ranked hypotheses. The goal is to precisely characterize the non-linear performance ‘ridge’ and determine whether its underlying mechanism is a ‘kinetic flux balance’ or a ‘hydrotropically-stabilized phase boundary’.

Multivariate Design Space

This experiment will employ a combinatorial strategy to explore the 4-dimensional design space of the key interacting components. A Sobol sequence sampling strategy will generate 96 unique conditions to efficiently explore the space and map interactions, explicitly avoiding the pitfalls of One-Factor-at-a-Time (OFAT) testing.

- **[Magnesium acetate] (mM)**: Range `4.0` to `25.0`
- **[Creatine phosphate] (mM)**: Range `10.0` to `60.0`
- **[ATP] (mM)**: Range `1.0` to `3.7`
- **[GTP] (mM)**: Range `1.0` to `3.3`

All other PURE system components will be held at their baseline concentrations to isolate the effects of the four chosen variables.

Batch Protocol Steps

1. **Stock Preparation**: Prepare concentrated, calibrated stock solutions for Magnesium acetate, Creatine phosphate, ATP, and

GTP.
2. **Master Mix Formulation**: Prepare a master mix containing all constant PURE system components, leaving volumetric headroom for the four variables.
3. **Combinatorial Plating**: Utilize an automated liquid handler to dispense the master mix and the precise, varying volumes of the four stock solutions into a 96-well plate according to the Sobol sequence design.
4. **Reaction Incubation**: Initiate reactions by adding a standard GFP DNA template and incubate the plate in a plate reader at 37°C.
5. **Data Acquisition**: Monitor GFP fluorescence and 90-degree light scattering kinetically over a 4-hour period.

Optimization Metrics

- **Primary Metric (Yield)**: `sigmoid_steady_state` calculated from the kinetic fluorescence curve. This will be used to generate a 4D response surface map to visualize the performance ridge.
- **Secondary Metric (Kinetics)**: `sigmoid_rate (1/h)` will be used to test the 'kinetic flux balance' hypothesis, as kinetic mismatches should manifest as suboptimal reaction rates.
- **Tertiary Metric (Precipitation)**: Endpoint and kinetic light scattering data will test the 'phase boundary' hypothesis. A sharp increase in scattering correlated with a collapse in yield will provide strong evidence.
- **Feedback Loop**: The resulting multi-dimensional dataset will be fed back into the surrogate model. This will not only test the current hypotheses but also dramatically improve the model's predictive accuracy within this critical subspace, enabling it to actively guide future optimization efforts directly toward the identified high-performance operational ridge.

References

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Dead Ends

- Hypothesis of Codon-Modulated Metabolic Optima: This hypothesis was consistently outcompeted in debates because it focused on a secondary phenomenon (protein-specific optimization) rather than providing a foundational explanation for the primary empirical evidence—the existence of a non-linear performance ridge for core metabolic components. The winning hypotheses were deemed more fundamental and parsimonious at this stage of research.